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"""
Stock Portfolio Tracker
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Calculates total investment value based on user-
entered stock names
and quantities, using a hardcoded dictionary of
stock prices.
Results are displayed on screen and saved to a CSV
file.
"""

import csv
from datetime import datetime

# Hardcoded stock prices
STOCK_PRICES = {
    "AAPL": 180,
    "TSLA": 250,
    "GOOGL": 170,
    "MSFT": 420,
    "AMZN": 190
}

def get_portfolio_input():
    """Ask the user for stock names and quantities
    until they choose to stop."""
    portfolio = {}

    print("=== Stock Portfolio Tracker ===")
    print(f"Available stocks: {'',
    '.join(STOCK_PRICES.keys())}")
    print("Enter 'done' as the stock name when
    you're finished.\n")

    while True:

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        stock = input("Enter stock name:
").strip().upper()

        if stock == "DONE":
            break

        if stock not in STOCK_PRICES:
            print(f"⚠️ '{stock}' is not a
recognized stock. Please try again.\n")
            continue

        qty_input = input(f"Enter quantity of
{stock}: ").strip()

        try:
            quantity = int(qty_input)
            if quantity <= 0:
                print("⚠️ Quantity must be a
positive whole number.\n")
                continue
            except ValueError:
                print("⚠️ Invalid quantity. Please
enter a whole number.\n")
                continue

        portfolio[stock] = portfolio.get(stock, 0)
        + quantity
        print(f"Added {quantity} shares of
{stock}.\n")

    return portfolio

def calculate_investment(portfolio):
    """Calculate per-stock value and total
investment value."""
    results = []
    total = 0

    for stock, quantity in portfolio.items():

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        price = STOCK_PRICES[stock]
        value = price * quantity
        total += value
        results.append({
            "Stock": stock,
            "Price": price,
            "Quantity": quantity,
            "Value": value
        })

    return results, total

def display_results(results, total):
    """Print a formatted summary of the
    portfolio."""
    print("\n=== Portfolio Summary ===")
    print(f"{'Stock':<8}{'Price':<10}{'Quantity':<10}{'Value':<10}")
    print("-" * 38)

    for row in results:
        print(f"{row['Stock']:<8}{row['Price']:<10}{row['Quantity']:<10}{row['Value']:<10}")

    print("-" * 38)
    print(f"Total Investment Value: ${total:,.2f}")

def save_to_csv(results, total,
filename="portfolio_summary.csv"):
    """Save the portfolio results to a CSV file."""
    with open(filename, mode="w", newline="") as
file:
        writer = csv.writer(file)
        writer.writerow(["Stock", "Price",
"Quantity", "Value"])

        for row in results:
            writer.writerow([row["Stock"],

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row["Price"], row["Quantity"], row["Value"]])

        writer.writerow([])
        writer.writerow(["Total", "", "", total])
        writer.writerow(["Generated on",
datetime.now().strftime("%Y-%m-%d %H:%M:%S")])

        print(f"\n✅ Results saved to '{filename}'")

def main():
    portfolio = get_portfolio_input()

    if not portfolio:
        print("No valid stocks entered. Exiting.")
        return

    results, total =
calculate_investment(portfolio)
display_results(results, total)
save_to_csv(results, total)

if __name__ == "__main__":
    main()

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